WORKSHOP

On computer vision for image segmentation

(and binary classification)

Vladimir PIMONOV, PhD (2026)

OUTLINE

SESSION 01

DATA PREPARATION

- Image Analysis
- Data Annotation
- Dataset Preparation

Practical work 01

- Data augmentations
- Data loading

Practical work 02

SESSION 02

MODEL TRAINING

- Model Architecture
- Training Pipeline
- Training Evaluation

Practical work 03

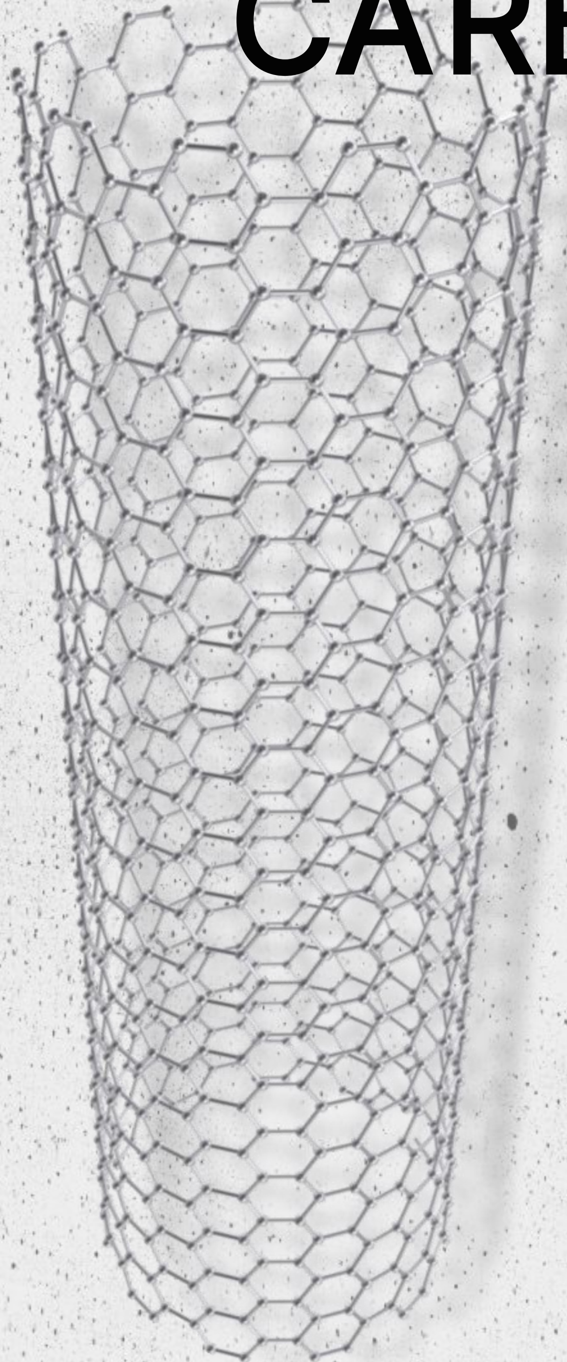
- Inference
- Post processing

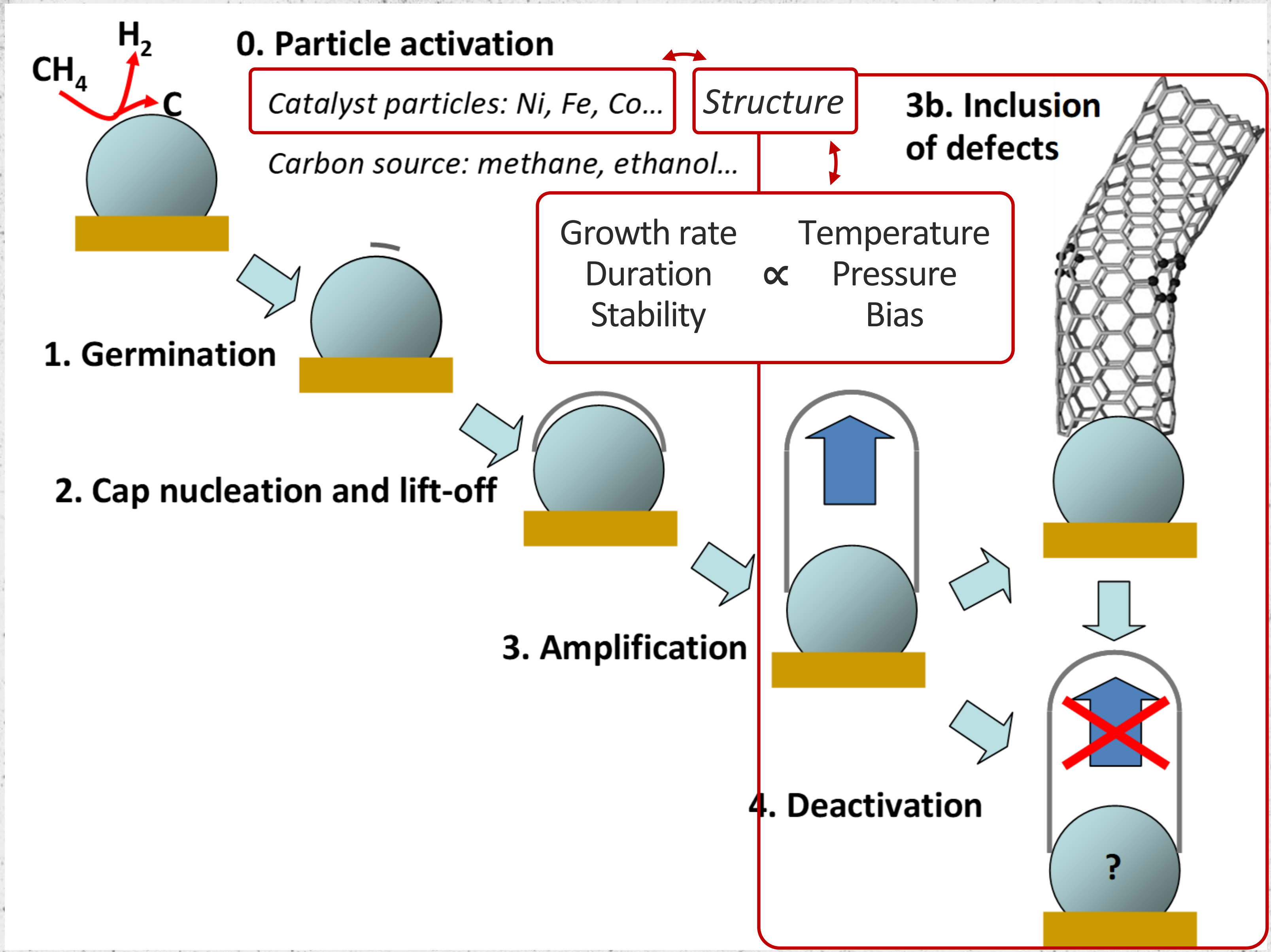
Practical work 04

INTRODUCTION

How did it started?

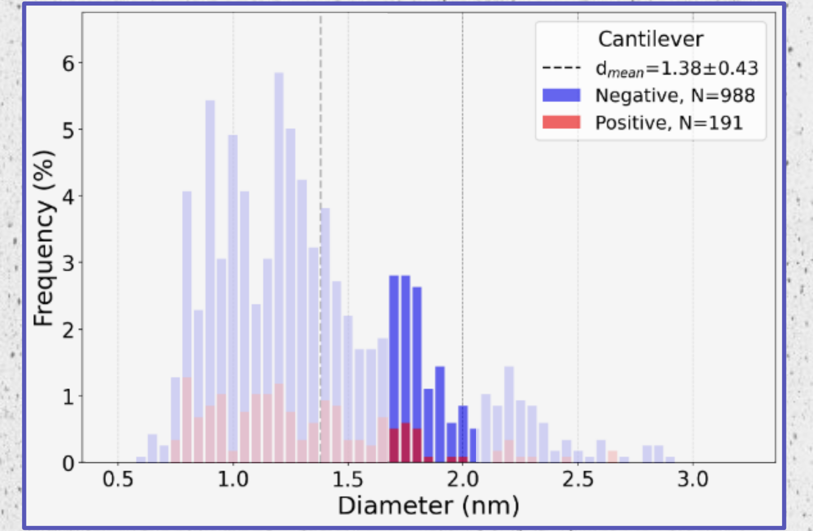
CARBON NANOTUBES GROWTH



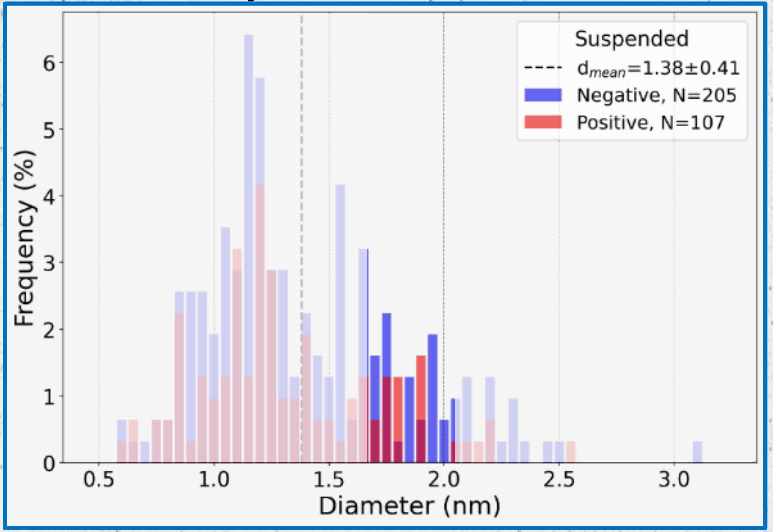


CARBON NANOTUBES STRUCTURE

Non-oriented CNTs

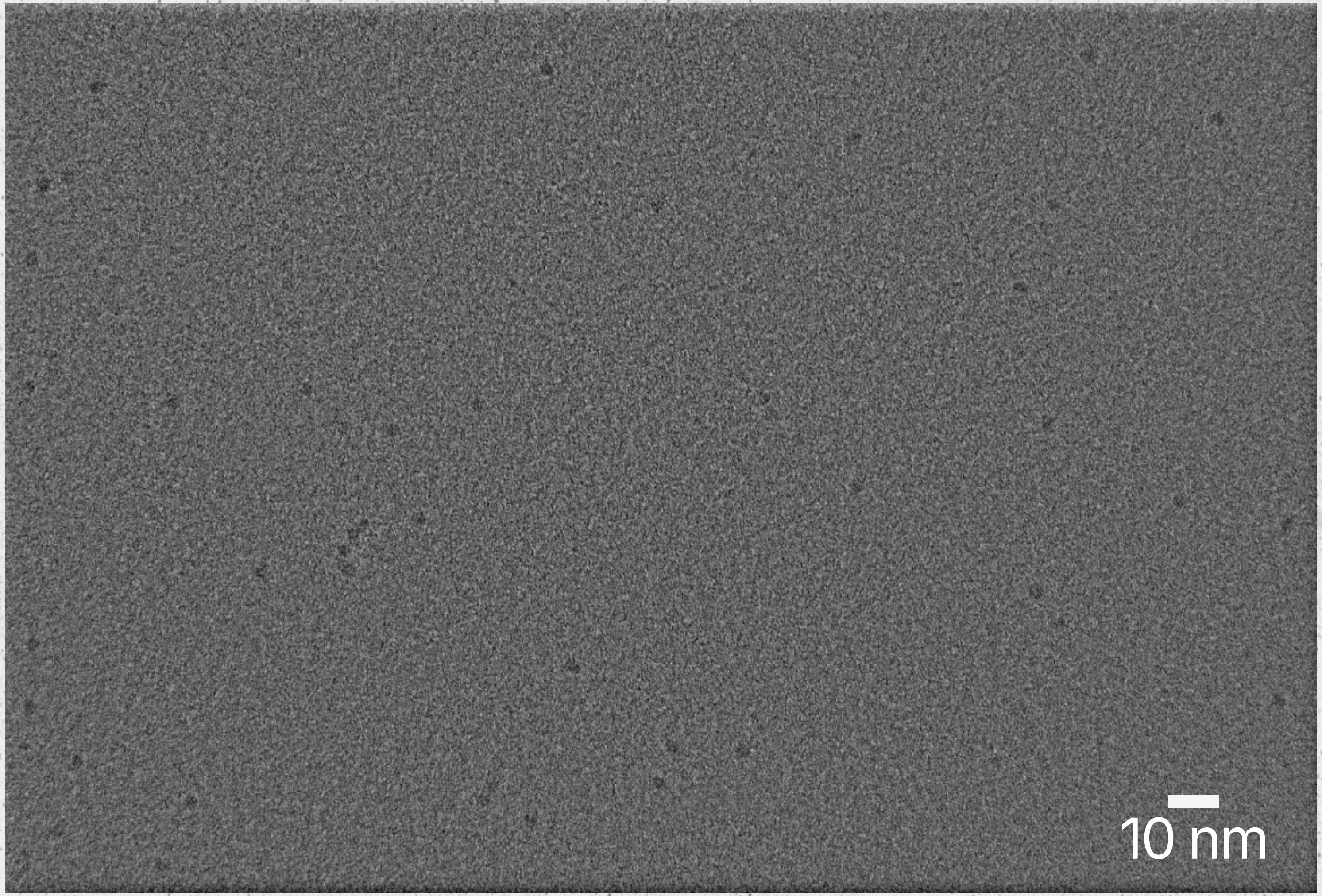


Suspended CNTs



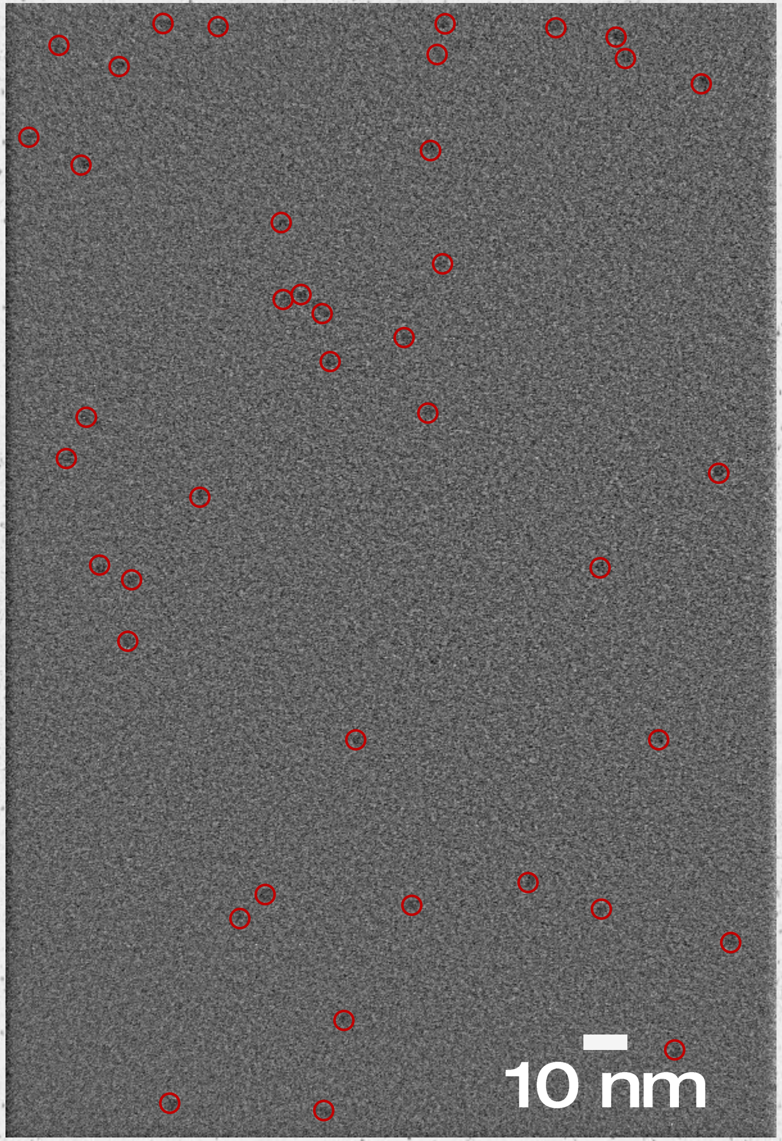
Mass selected catalyst clusters

To narrow down the nanotubes diameter distribution

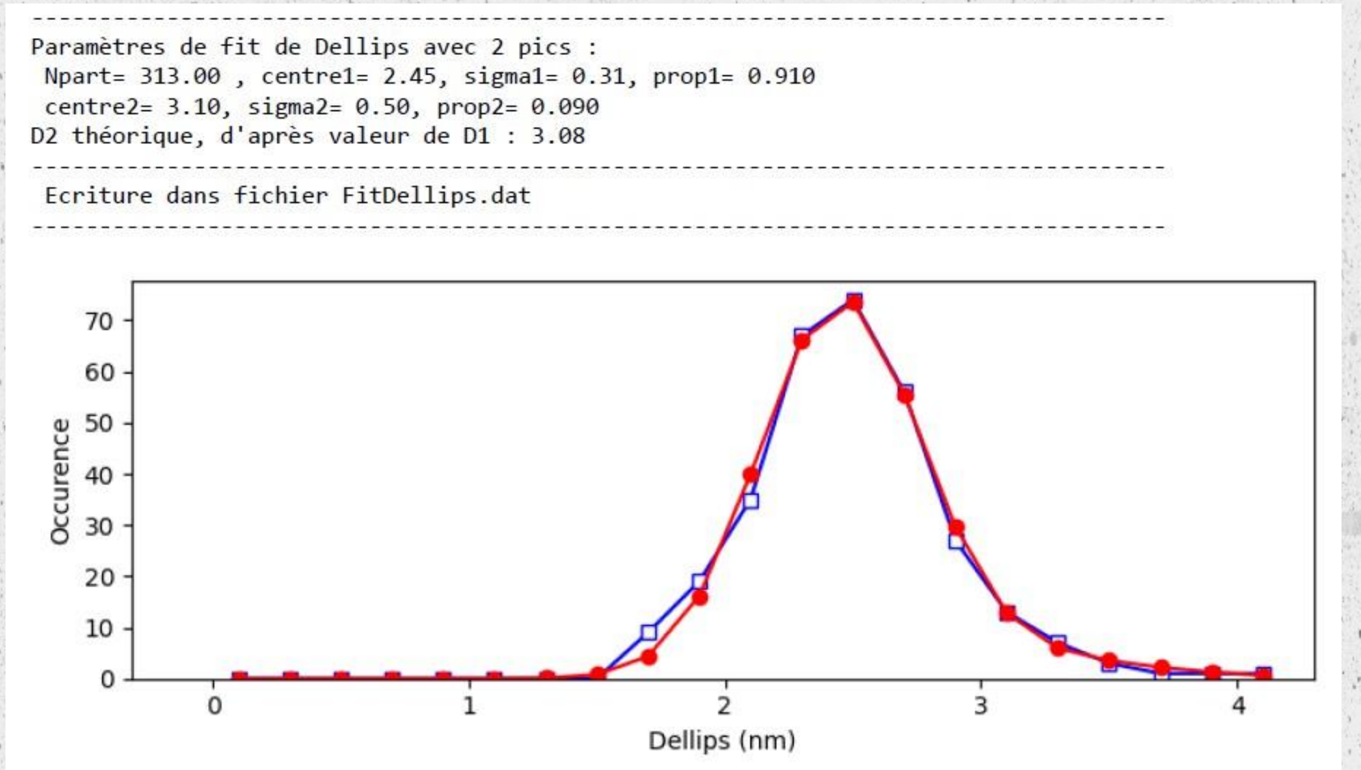


CATALYST EVALUATION

To understand catalyst size effect on CNTs diameter it is necessary to estimate its own diameters



HOW?



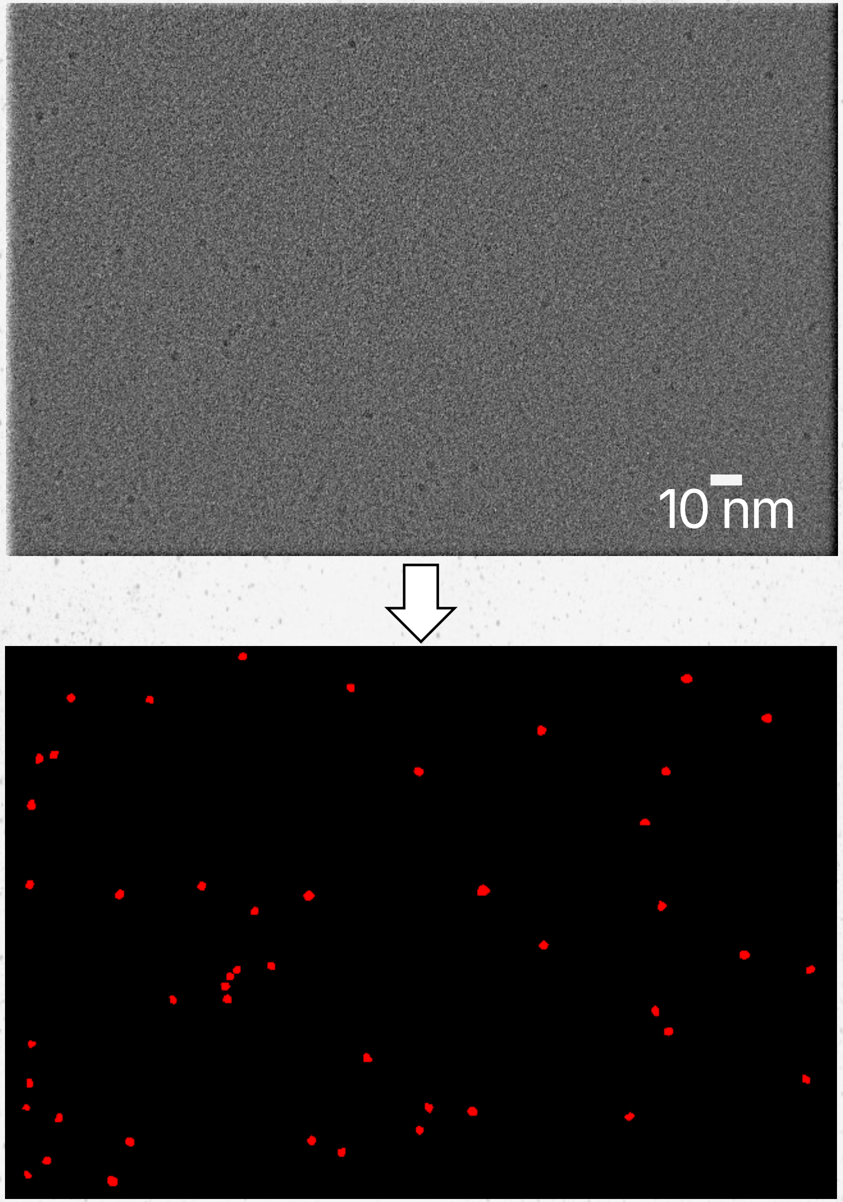
Credit: Florent Tornus

PRACTICAL WORK 01

Image Analysis
Data Annotation
Dataset Preparation

DATA ANNOTATION

Image

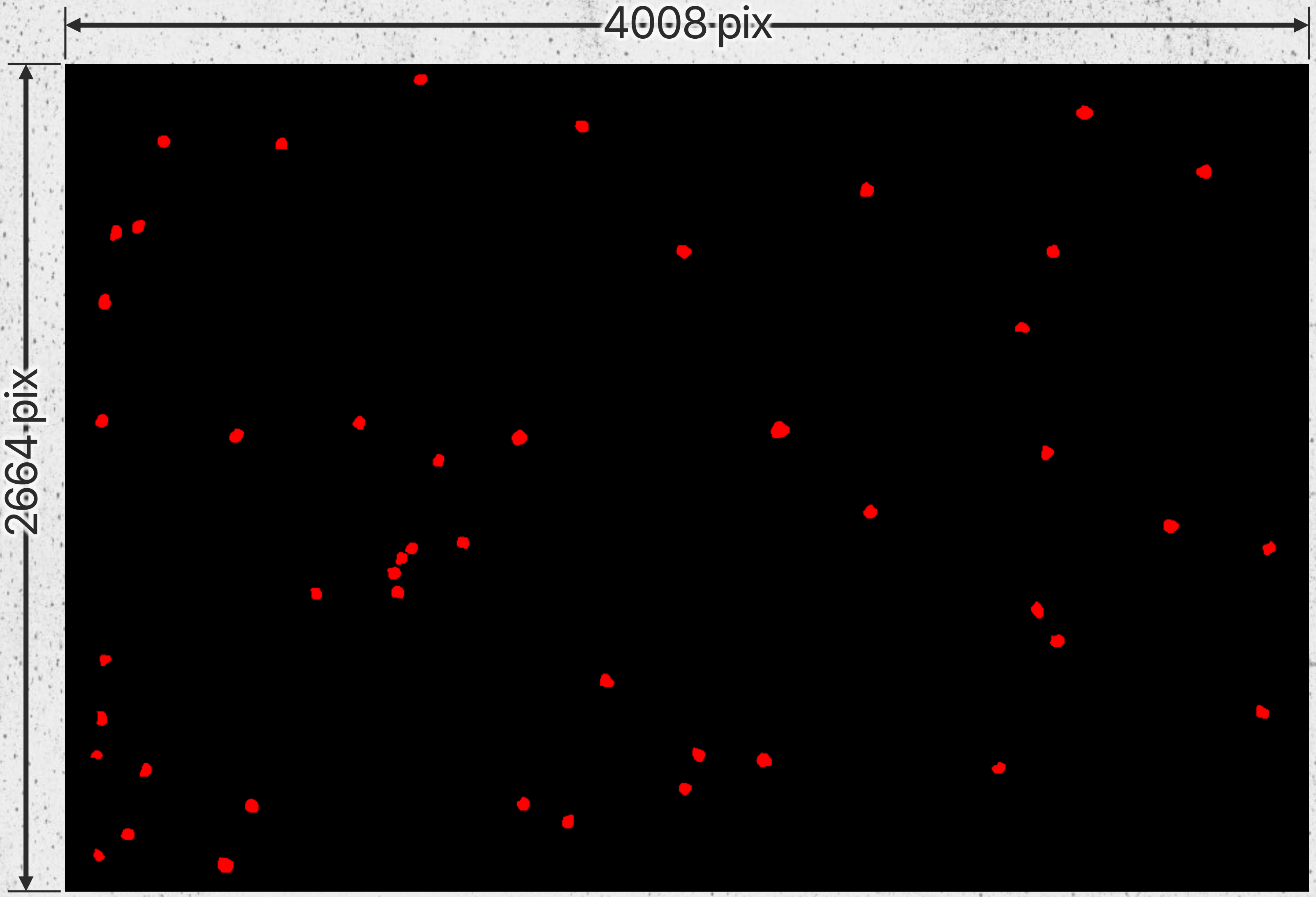


Mask

Method	+	-
Manual: Paint; Photoshop, GIMP	Low entry barrier Fine control High precision	Time consuming Human error Low scalability
Semi-automated: LabelMe; ImageJ	Faster workflow Human supervision Better consistency	Risk of bias Tools required Correction needed
Automatic: Custom built models	High throughput Scalability Minimal control	Labeled data required Model tuning needed

MANUAL DATA ANNOTATION

IMAGE ANALYSIS



Size: 4008x2664 pix

- Too large to use the entire image for training
- Prepare patches

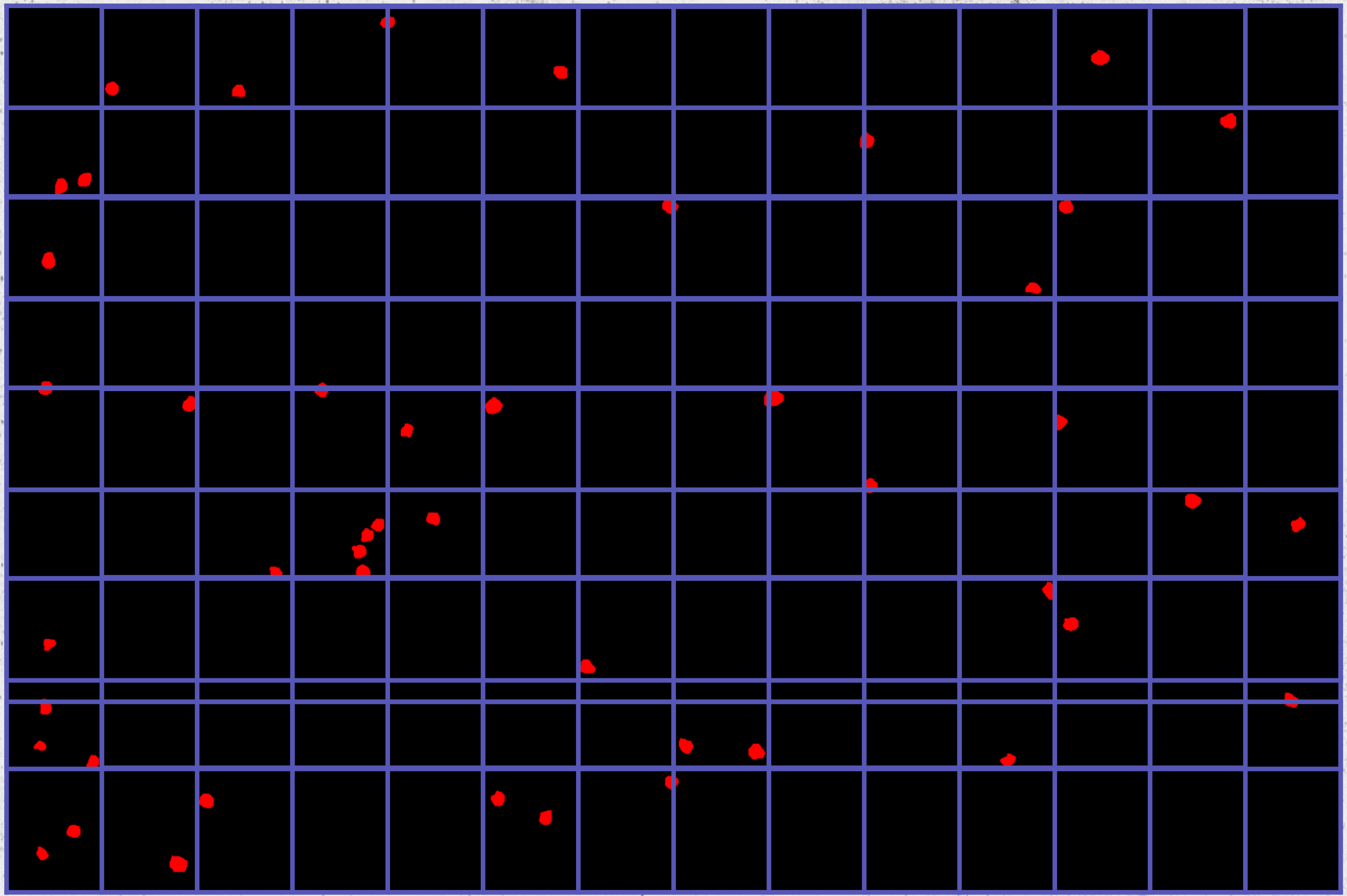
Nanoparticles number:

- ~50 per image
- Scarce and sparse positive examples
- Limit number of empty patches

Nanoparticles diameter:

- ~50 pix (3 nm)
- Convolution may shrink nanoparticles to zero
- Limit maximum size of patches

DATASET PREPARATION



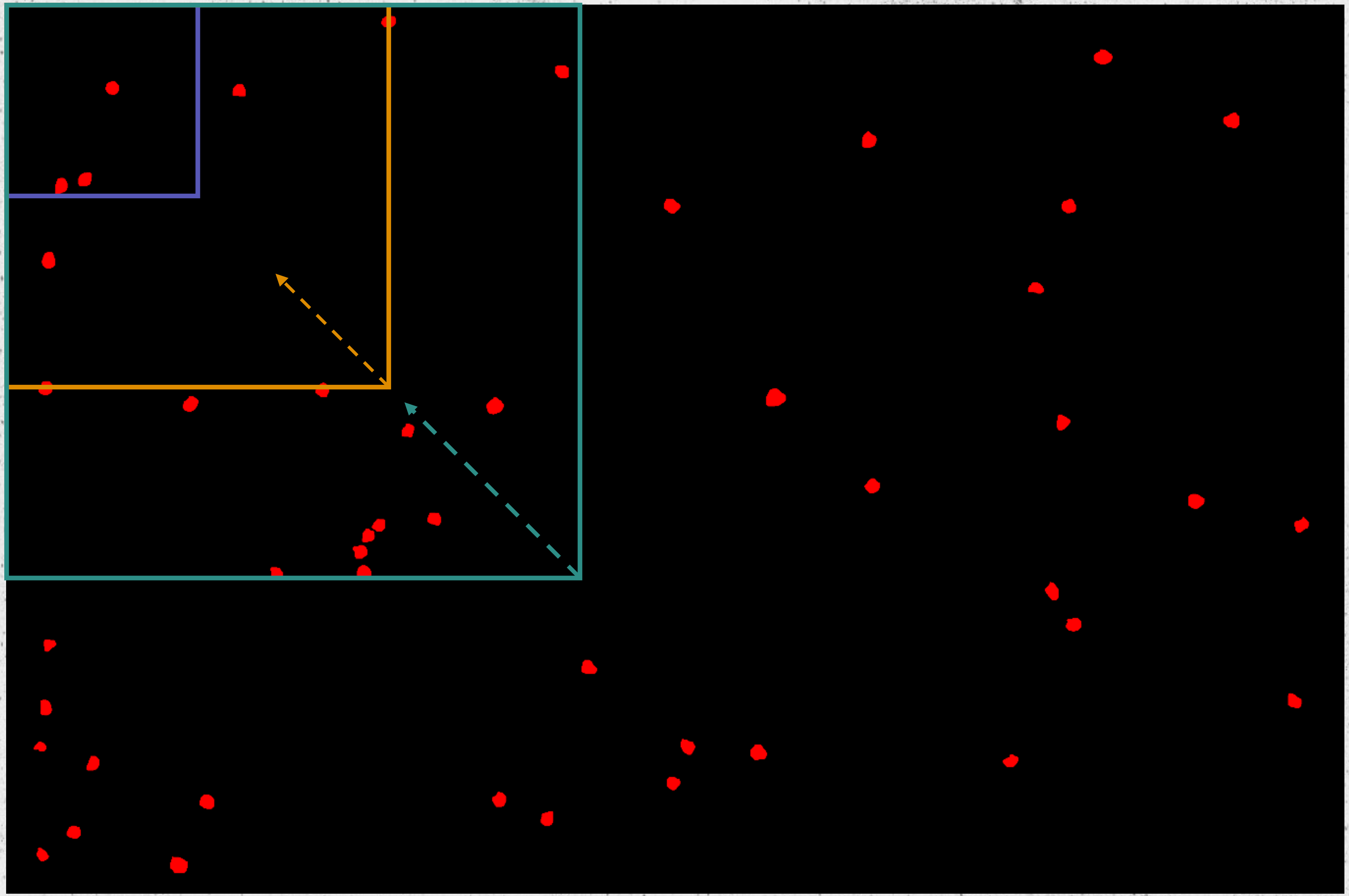
Prepare patches

- Limit memory use
- Expand dataset

With overlaps

- Further expand dataset
- Help learn border cases

DATASET PREPARATION



- Prepare patches
  - Limit memory use
  - Expand dataset
- With overlaps
  - Further expand dataset
  - Help learn border cases
- Larger patches
  - Further expand dataset
  - Learn larger context
  - Equivalent of different magnification
- Resize all patches to base size
  - Limit memory use
  - Boost training efficiency

DATASET PREPARATION

Prepare patches with overlaps

Use larger patches

Resize all patches to base size

NB! Avoid instant patch generation to reduce amount of unused data

1. Calculate

Patch locations:
  x, y

Nanoparticle characteristics:
  quantity, locations

Patch parameters:
  empty, almost empty

2. Select data

Limit the instances:
  empty, almost empty

Split data:
  train, validation, test

3. Generate data

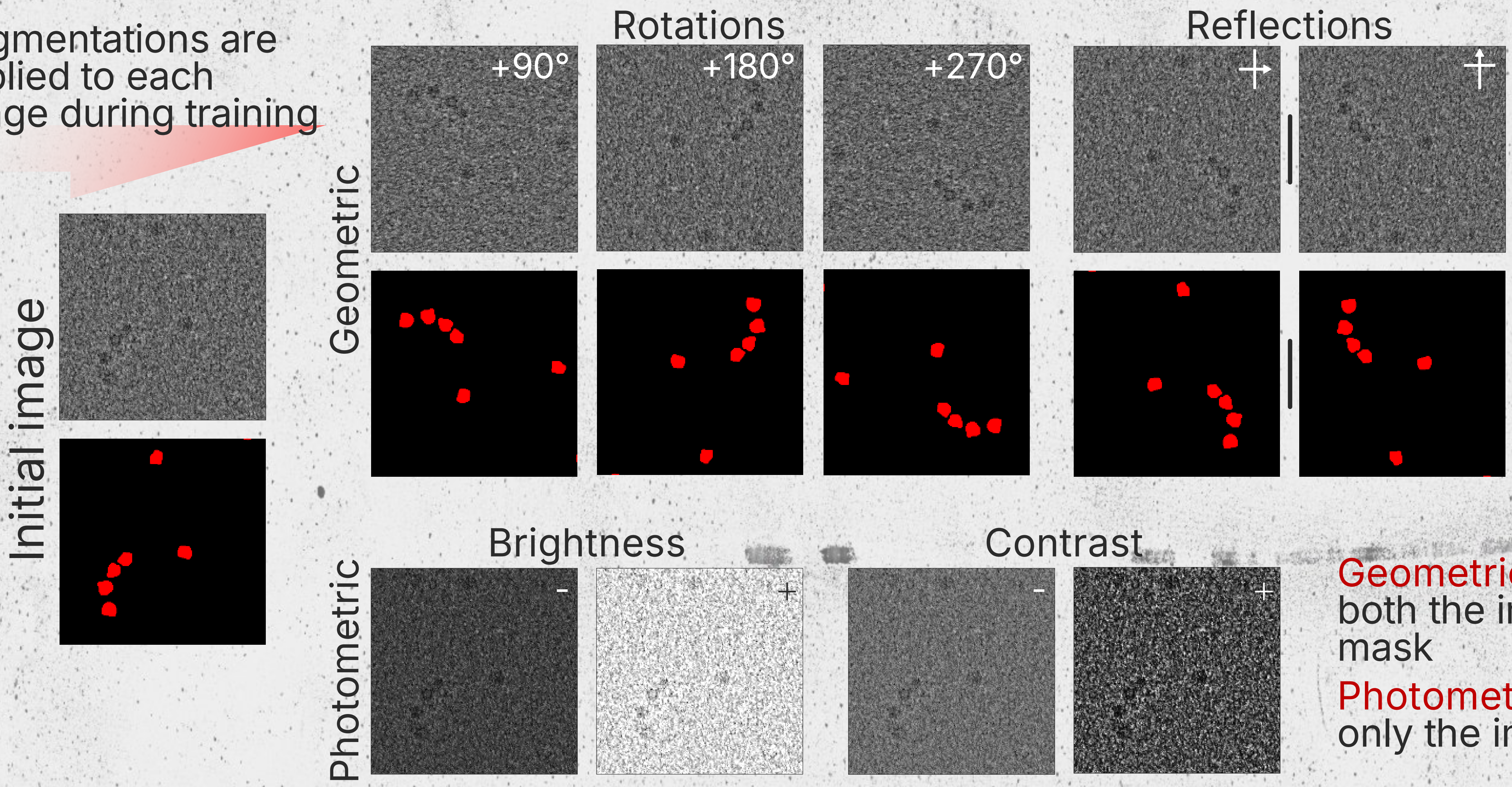
Prepare patches
  crop and save image and mask patches

PRACTICAL WORK 02

Data augmentations
Data loading

DATA AUGMENTATIONS

Augmentations are applied to each image during training



Geometric affects both the image and a mask

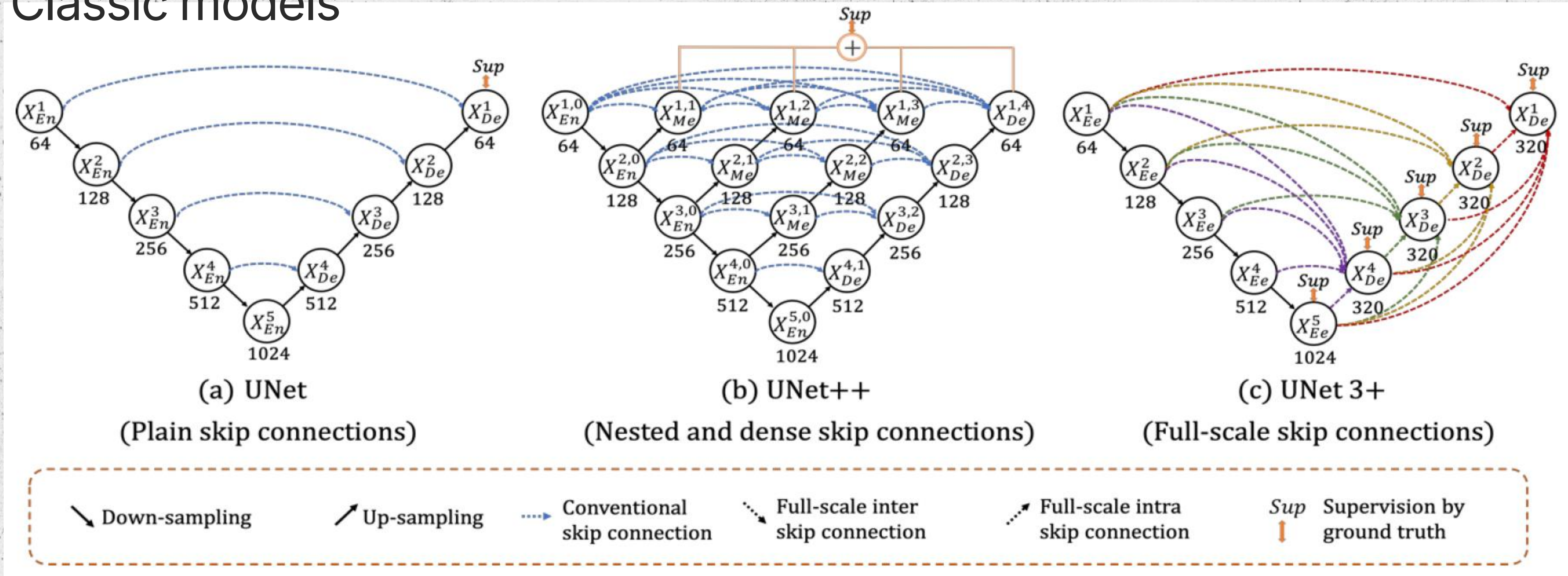
Photometric affects only the image

PRACTICAL WORK 03

Model Architecture
Training Pipeline
Training Evaluation

MODEL ARCHITECTURES

Classic models



Encoder-decoder model with skip-connections

Model with nested skip pathways connecting features between the low- and high-level layers

Full scale feature fusion model, at each decoder level aggregates information from all encoders

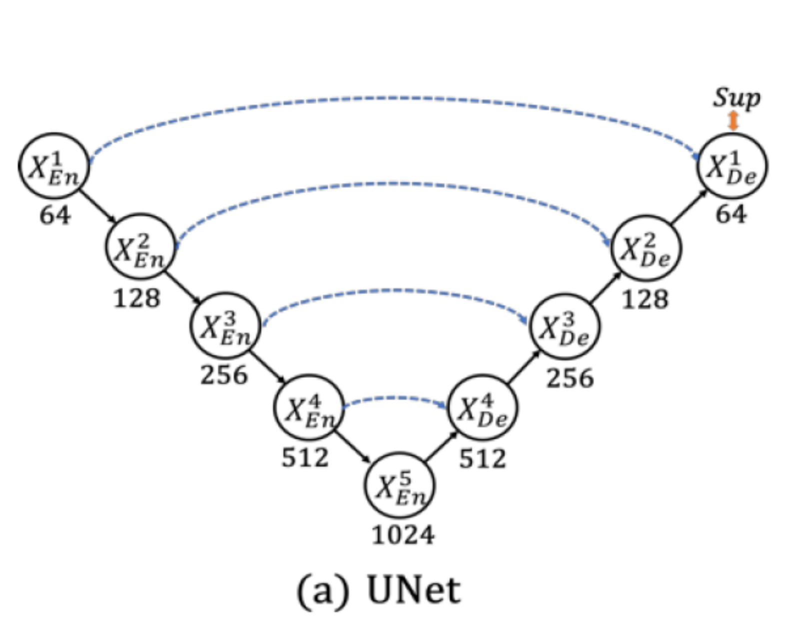
Ronneberger, O., et al. *MICCAI 2015*, 9351, (2015)

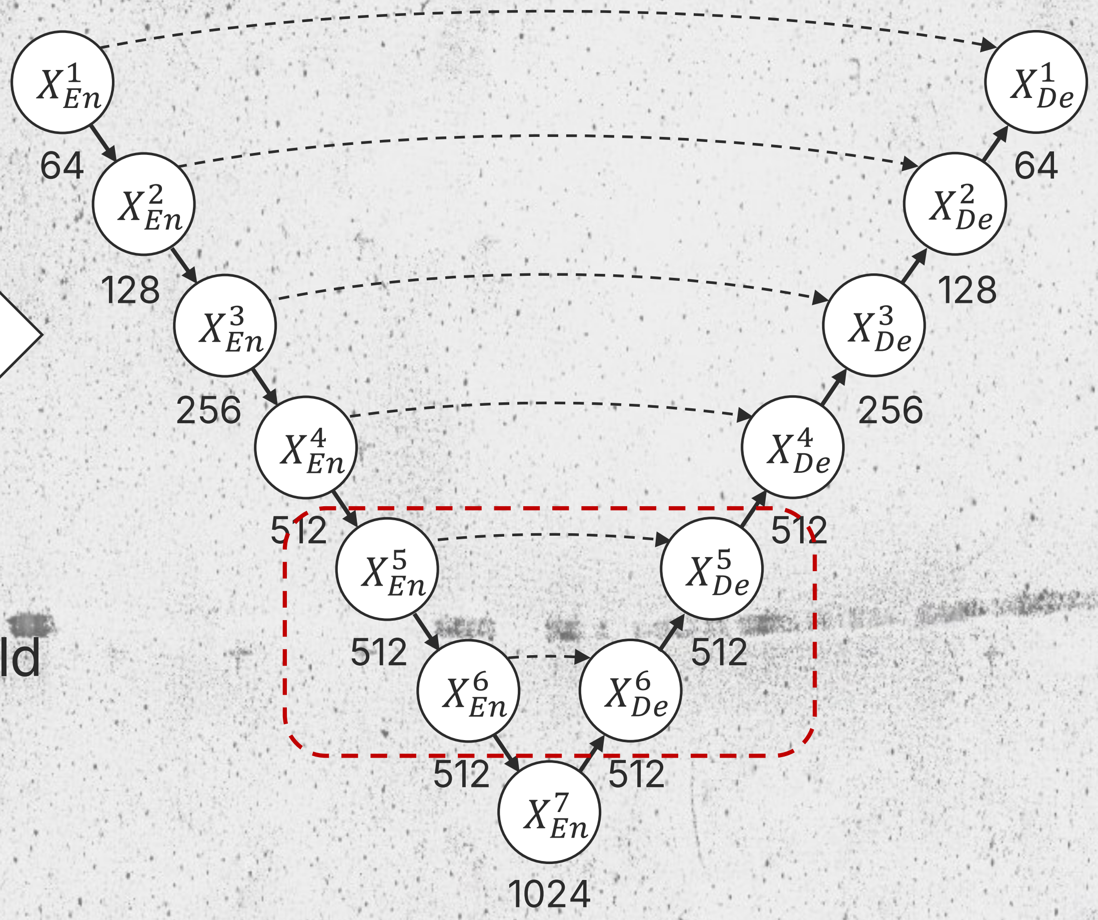
Zhou, Z., et al. *DLMIA ML-CDS 2018* (2018)

H. Huang et al., *ICASSP 2020* (2020)

MODEL ARCHITECTURES

Deep models





Additional encoder-decoder layers further increase the receptive field of the model

NB! To avoid spatial collapse of convoluted images deep architectures require larger patches

TRAINING PIPELINE

1. Select model
2. Select loss functions
   What parameter will be used to evaluate the training
3. Select optimization algorithm
   Optimizer will update the model weights
   Scheduler will reduce learning rate during training
4. Select hyperparameters
   Number of epochs
   Learning rate / Weight decay
   Loss function parameters
   Optimizer parameters
   Scheduler parameters
5. Train the model
6. Evaluate the result

X. Use for inference

7. Fail and repeat

Binary cross entropy (BCE)

Punish wrong prediction of true class

BCE Loss = -∑

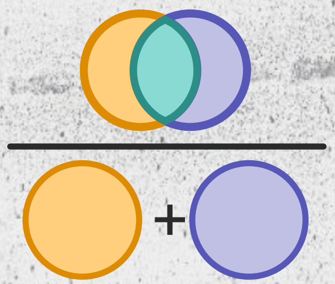
Weighted BCE

Add more weight to predictions of certain class in highly imbalanced datasets

Dice + BCE

Combine BCE with Dice to enhance boundaries detections

Dice = 2

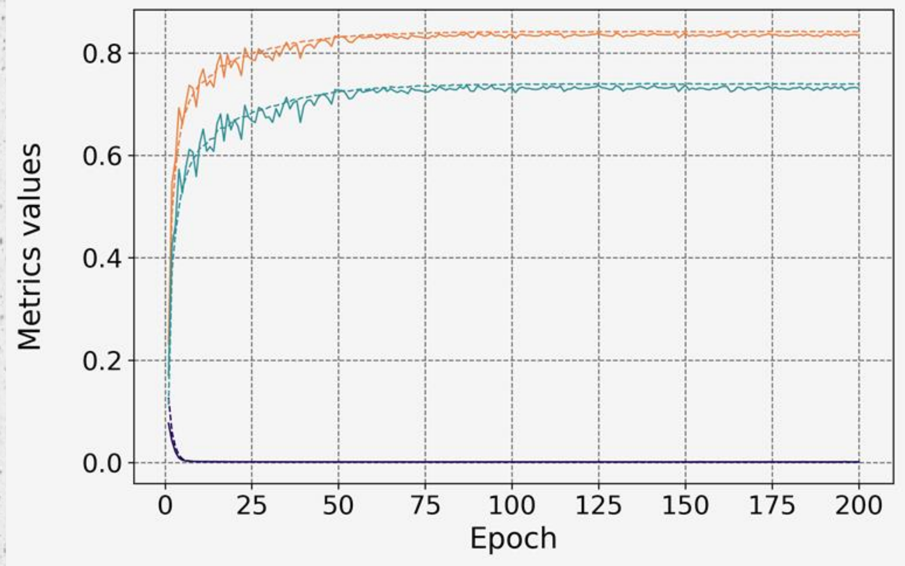


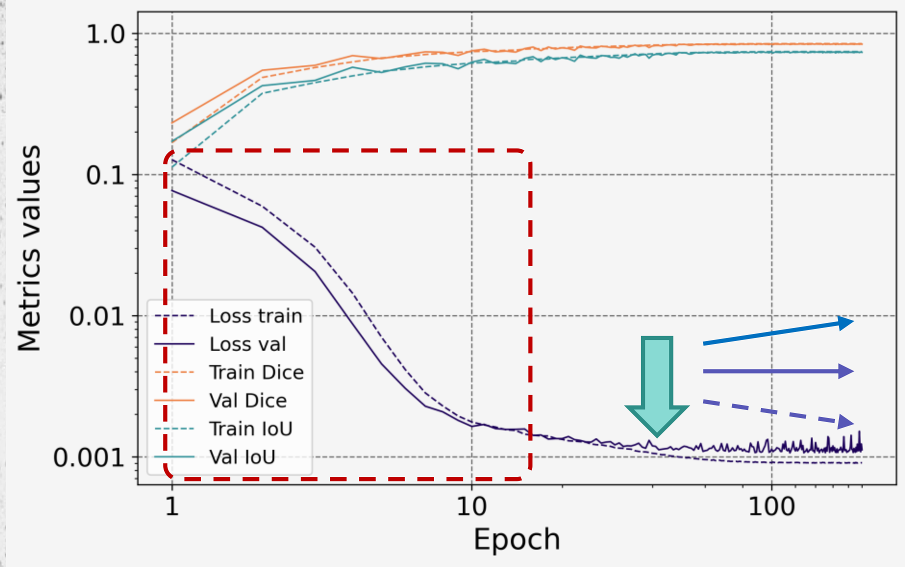
Focal loss

Increase importance of hard predictions to enhance boundaries detection

TRAINING EVALUATION

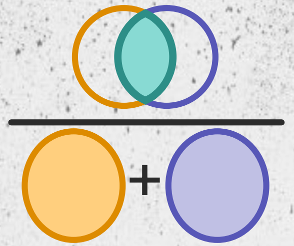
Training curves

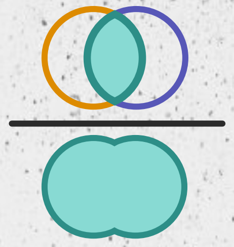




Dice: $[0, 1]$

Intersection over Union (IoU) $[0, 1]$

Dice = 2 

IoU = 

Dice and IoU measures the overlap between predicted and true masks
IoU is more sensitive to mismatches

Loss function: $(-\infty, 0]$

Comparing training and validation loss helps assess training quality.

Training, validation loss decreasing
Model is training, weights are improving

Training decreasing, validation loss plateaus
Model weights are optimal for given pipeline
Inflection point weights used as “early stop”

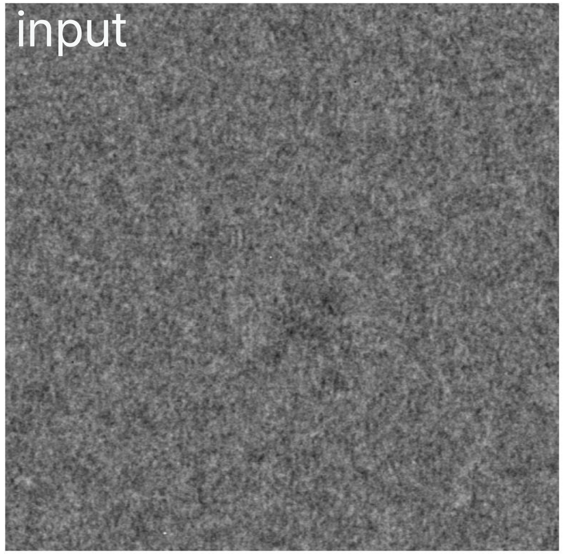
Training decreasing, validation loss grows
Model overfitting, predictions are failed

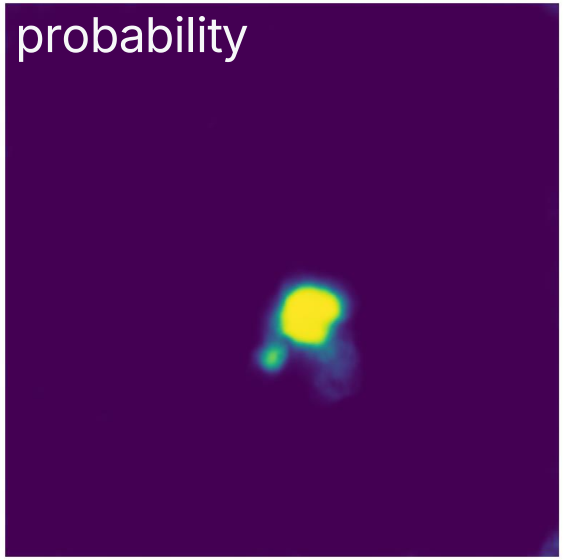
PRACTICAL WORK 04

Inference
Post processing

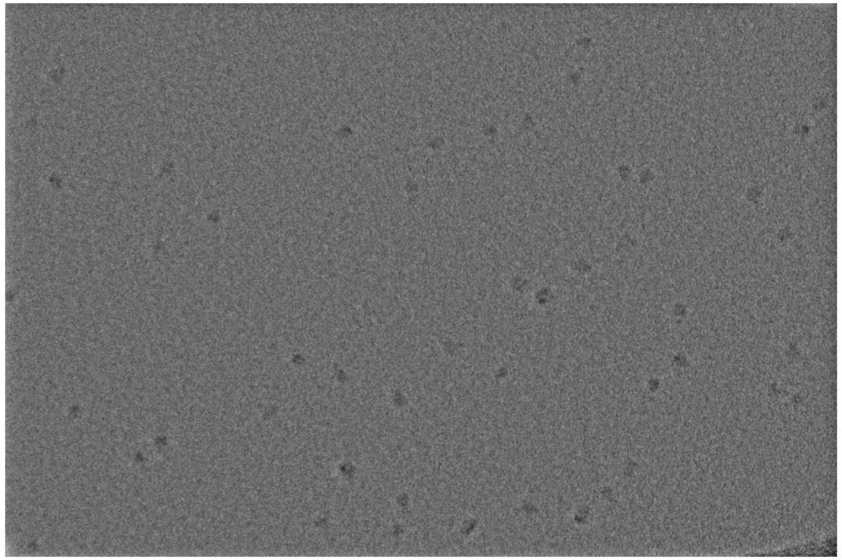
INFERENCE

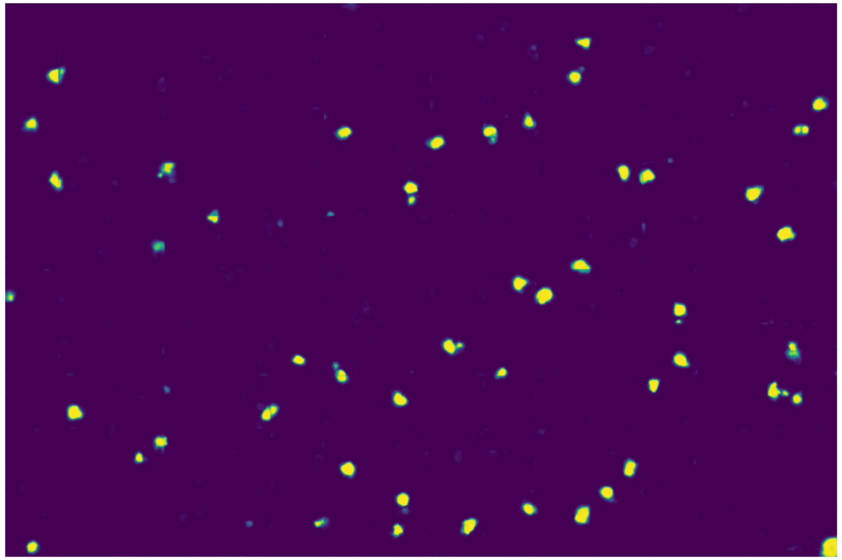
Infer patch





Tile the full image





Model output

Prediction is continuous confidence map with values normalized between 0 and 1.

It can be binarized by applying a chosen threshold.

Patch size

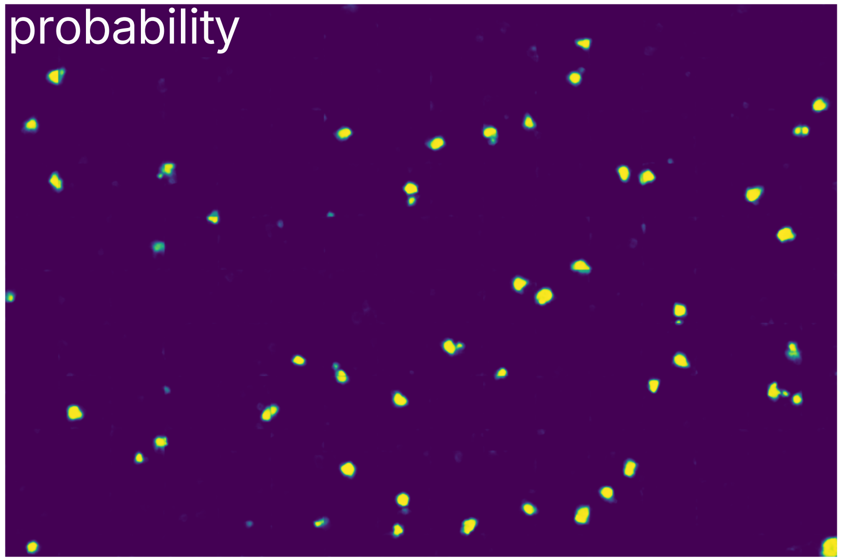
Inference patches do not need to have the same size as the training patches

Tiling with overlap

Predictions near patch borders are typically less reliable than those in the central region.

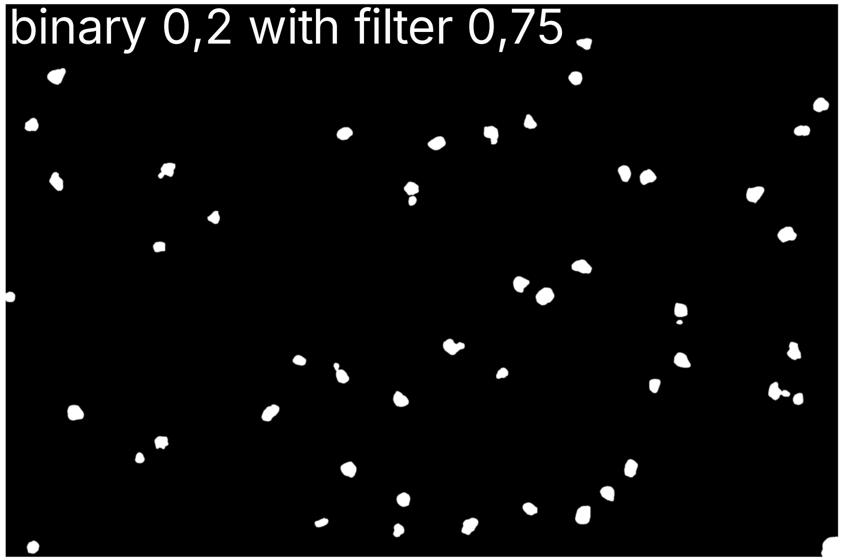
To reduce this effect, large images are inferred using overlapping patches.

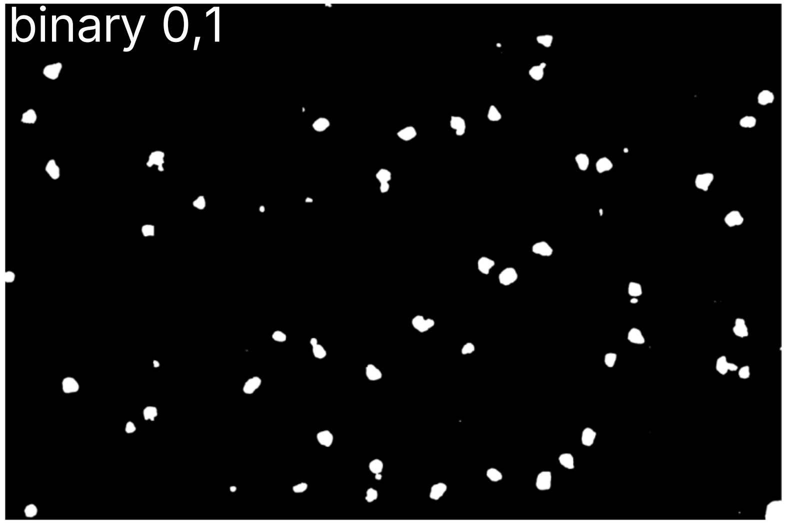
POSTPROCESSING

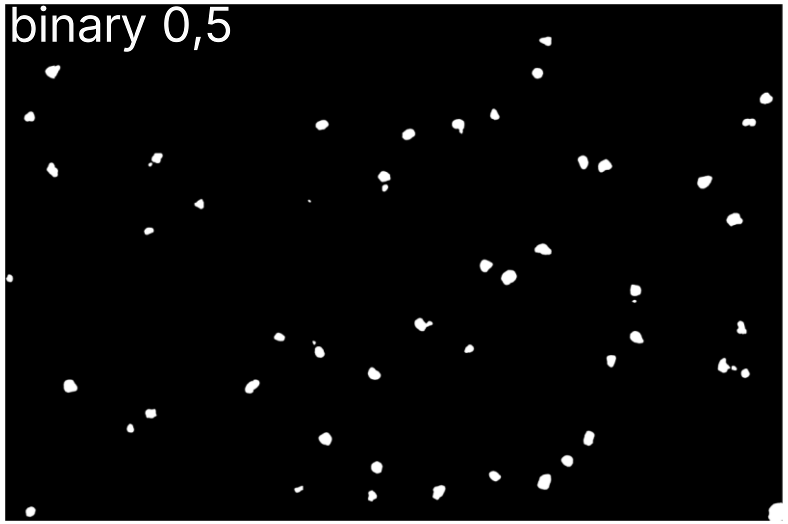


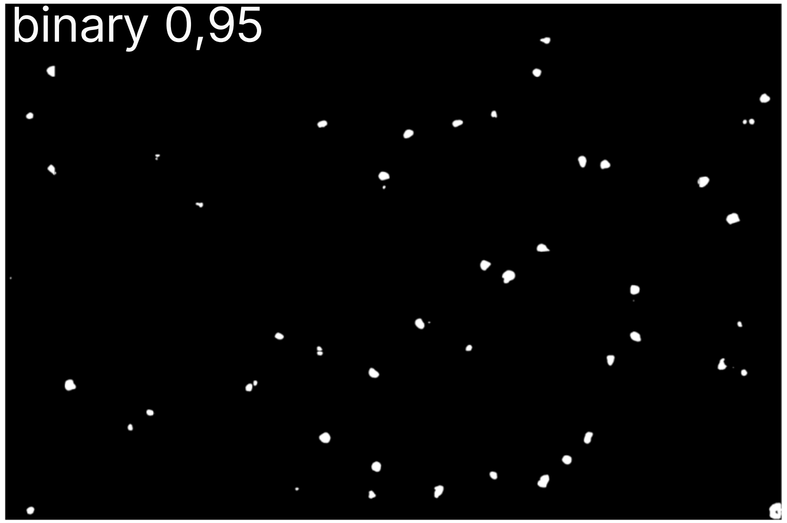
Different binary thresholds preserve different levels of detail.

High-confidence predictions can be used to filter low-confidence hallucinations.









Know your data before training the model

Dataset preparation is part of the model

Stronger architecture does not replace a clean pipeline

Inference is more than applying a threshold

WORKSHOP

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(and binary classification)

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